

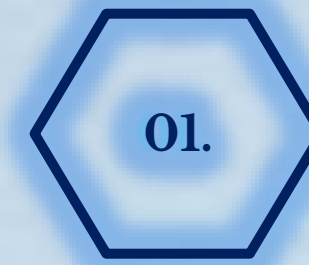


Predicting Diabetes from Patient-Level Health Indicators:

A Comparative Machine Learning Analysis
of the *Diabetes Health Indicators* Dataset

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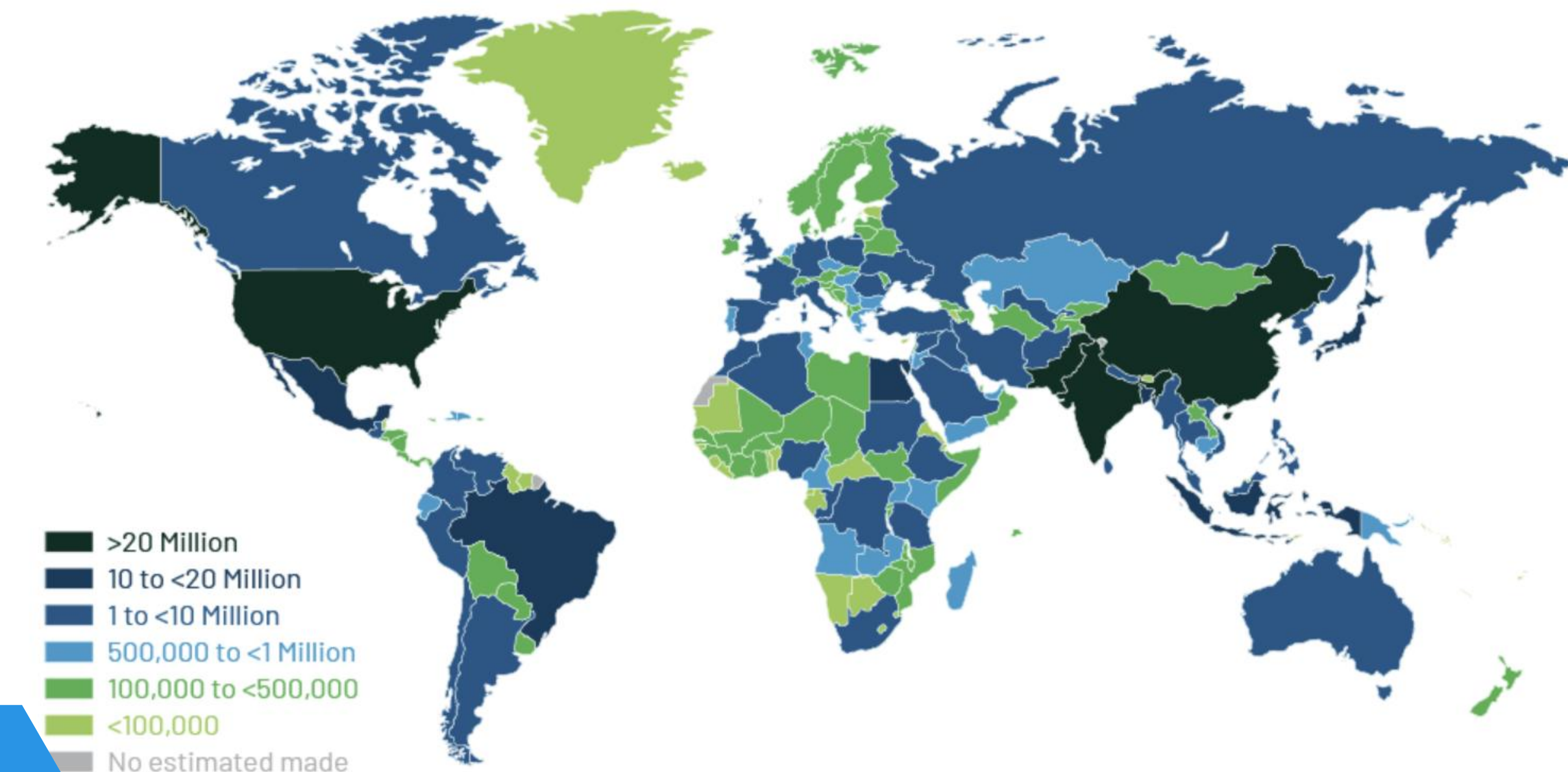
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Problem Motivation

Diabetes as a Global Health Challenge

Estimated Total Number of Adults (ages 20-79) With Diabetes in 2021



Diabetes affects hundreds of millions of adults worldwide.

Global prevalence continues to increase across regions.

Early detection is critical for reducing long-term complications.

Diabetes Types

Major Clinical Categories

- **Type I Diabetes:**
Autoimmune destruction of insulin-producing cells
- **Type II Diabetes:**
Insulin resistance and metabolic dysfunction
(*most common*)
- **Gestational Diabetes:**
Occurs during pregnancy
- **Prediabetes:**
Elevated glucose levels below the diagnostic threshold

Prediction Target (Binary Diagnosis)

1 (Yes):

Any diagnosed stage

(Type 1, Type 2, Gestational, Prediabetes)

0 (No):

No diabetes diagnosis



Motivation & Research Questions

- *Large health dataset* availability
- ML can *identify patterns* in this data
- Early detection for *prevention* and *treatment*

Research Questions

Primary:

- How do different supervised learning models compare for diabetes prediction?

Secondary:

- Which health indicators are most influential?
- What is the incremental predictive contribution of the diagnostic variable HbA1c?

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Dataset & Initial Screening

Diabetes Health Indicators Dataset

TARGET VARIABLES:

- Diagnoses
- ~~Stage~~
- ~~Risk Score~~

Lipid Profile:

Cholesterol Total
HDL Cholesterol
LDL Cholesterol
Triglycerides

Socioeconomic:

Education
Employment
Income

Demographic:

Ethnicity
Gender
Age

Behavioral:

Smoking Status
Alcohol Consumption
Physical Activity
Screen Times
Sleep Hours
Diet Score

Glycemic Markers:

Glucose Postprandial
Glucose Fasting
Insulin Level
HbA1c

Cardio-vascular:

Systolic BP
Diastolic BP
Heart Rate

Medical History:

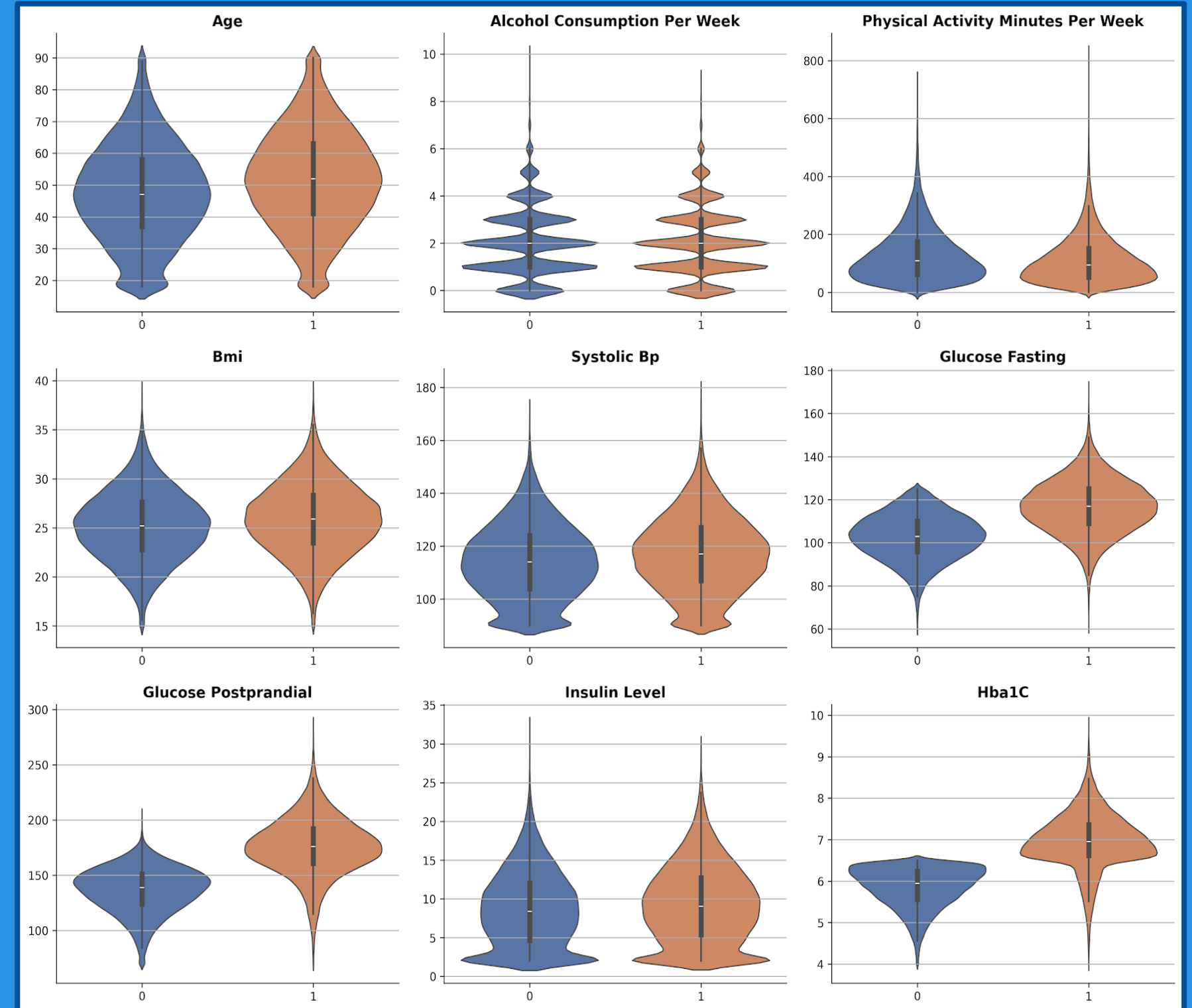
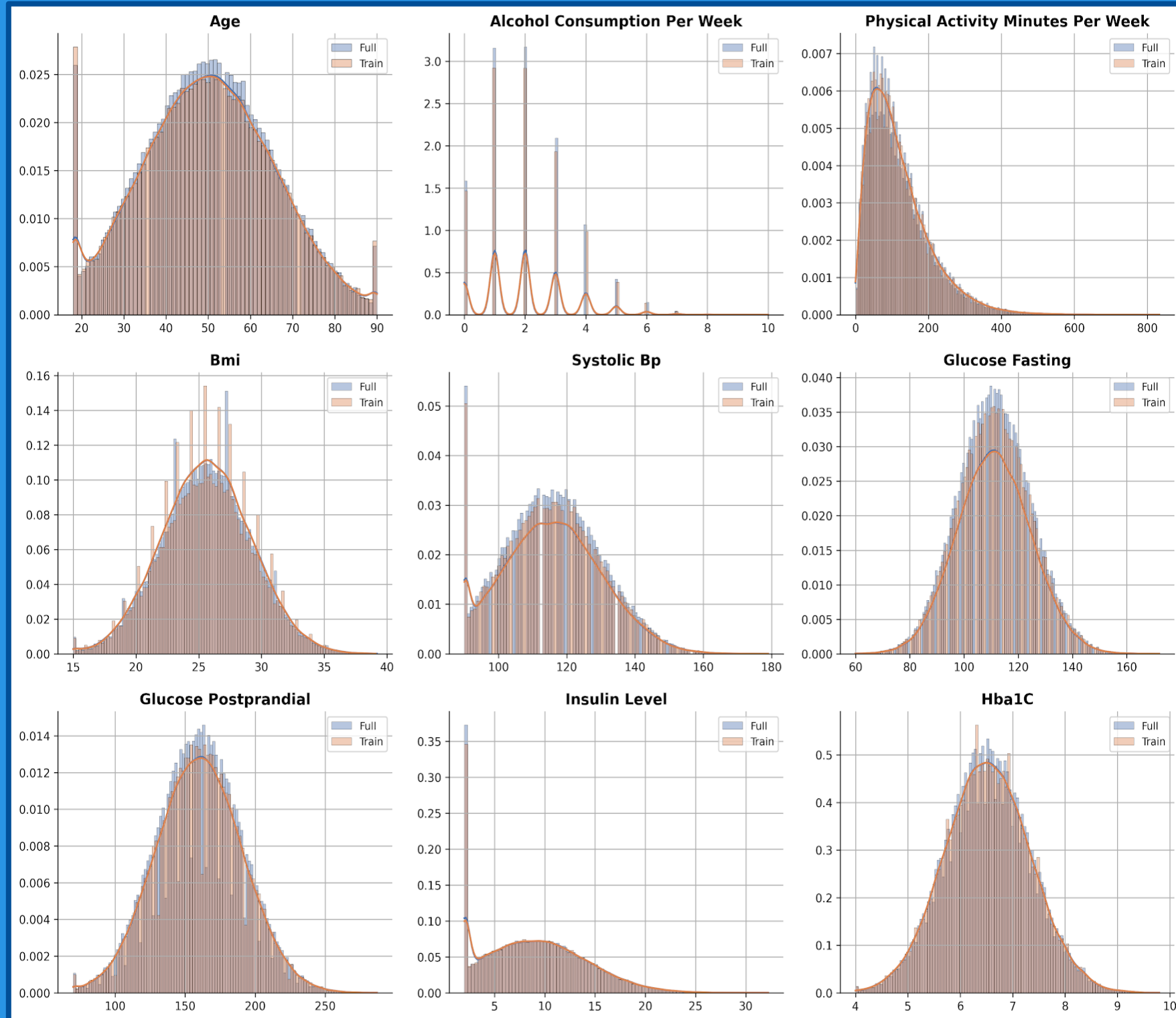
Family Diabetes
Cardiovascular
Hypertension

Anthro-pometric:

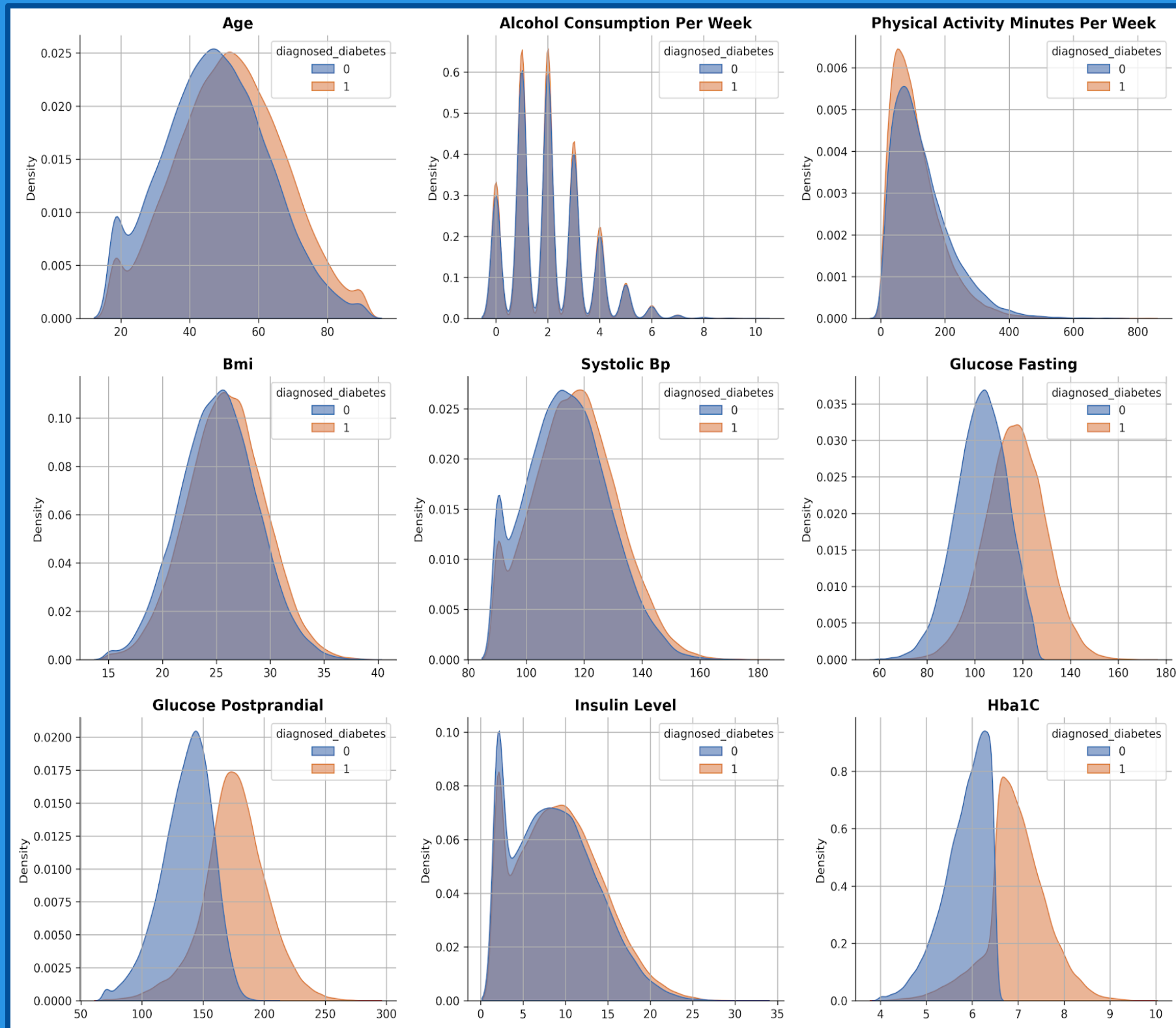
Waist-to-Hip-Ratio
BMI



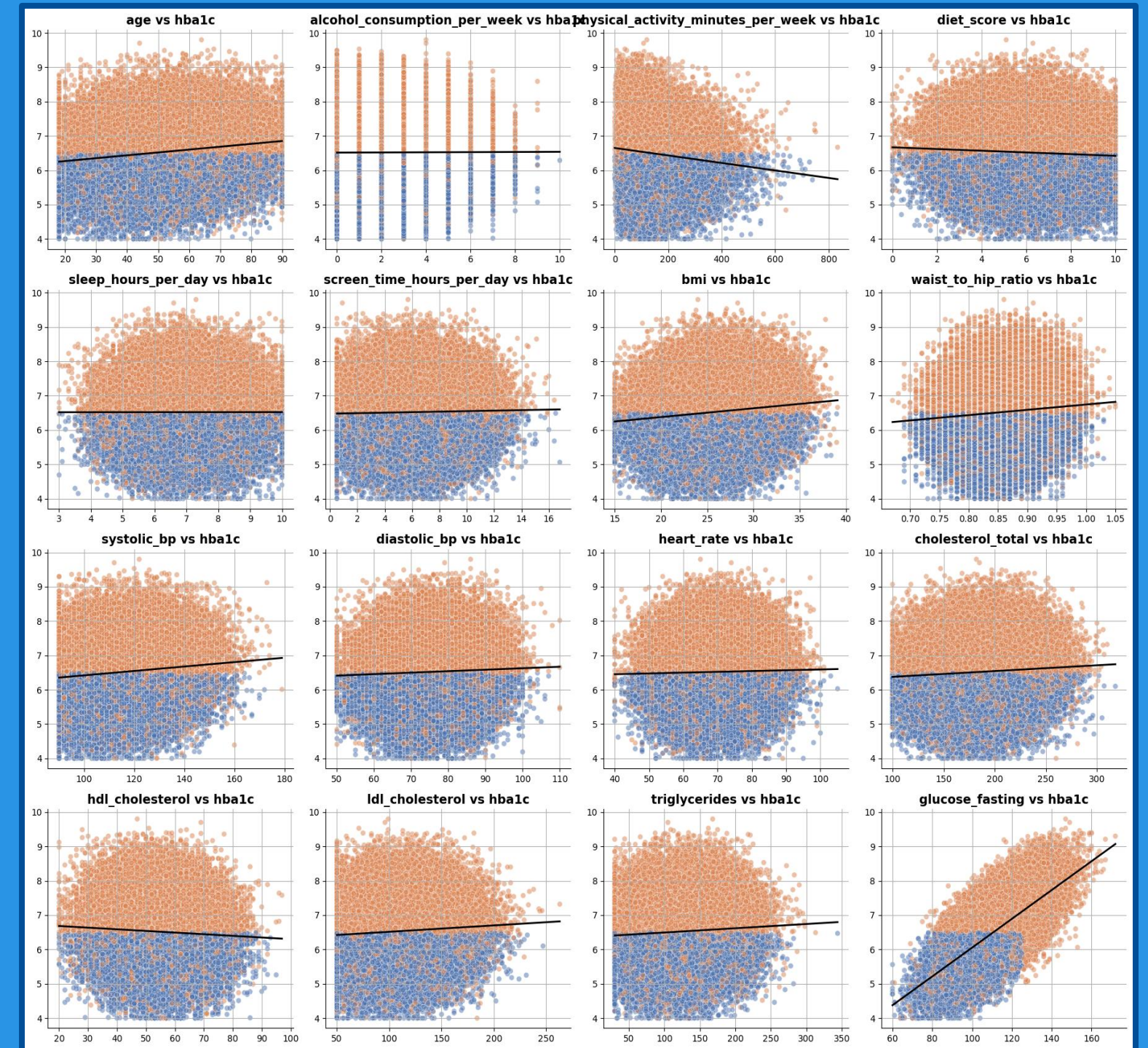
EDA: Distributions & Outliers



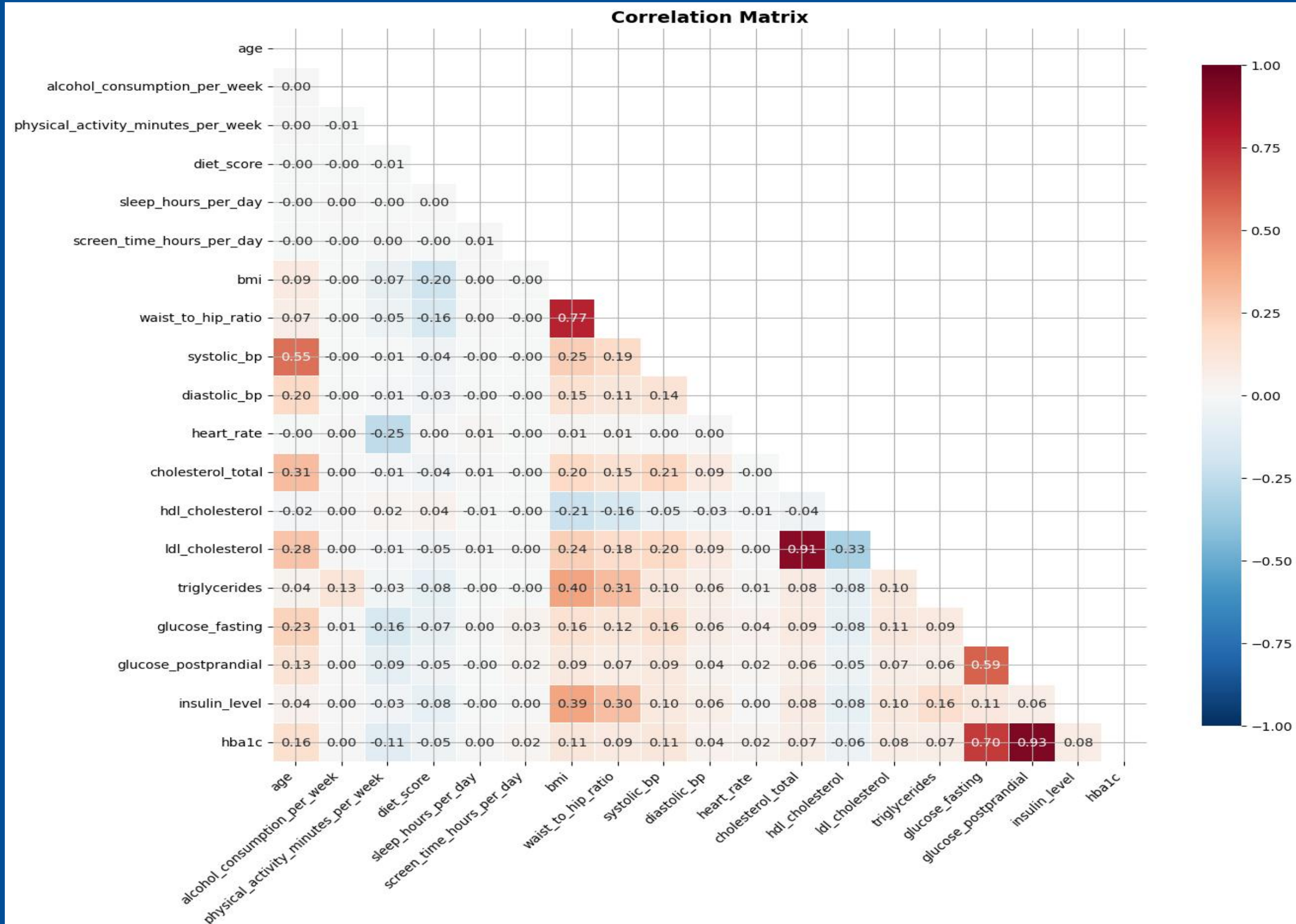
EDA: Nonlinear & Interactional Patterns



HbA1c vs Other Predictors



Correlation & Binning



Glycemic Markers:

HbA1c

~~Glucose Postprandial~~

~~Glucose Fasting~~

Insulin Level

Lipid Profile:

~~Cholesterol Total~~

HDL Cholesterol

LDL Cholesterol

Triglycerides

Anthropometric:

BMI

~~Waist to Hip Ratio~~

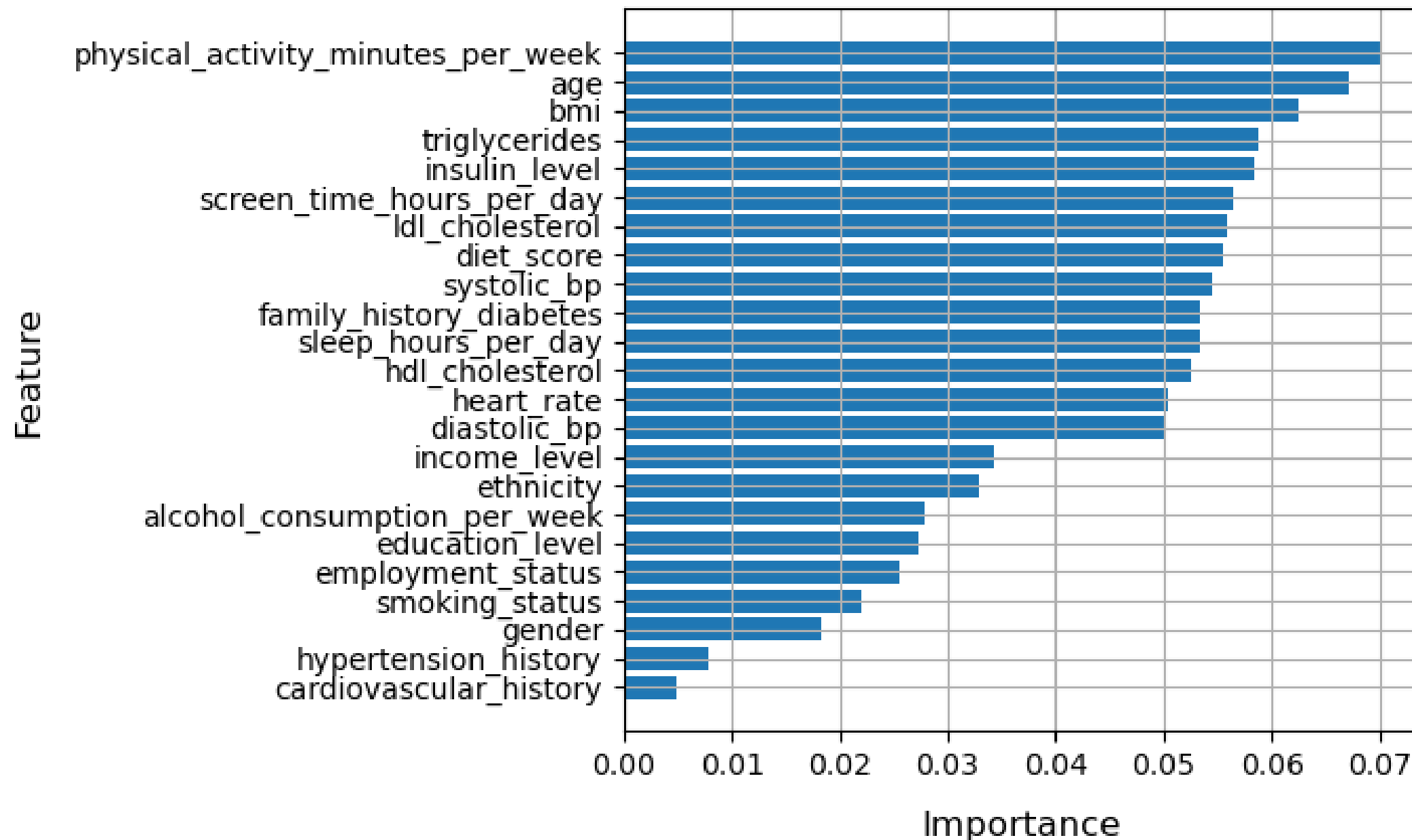
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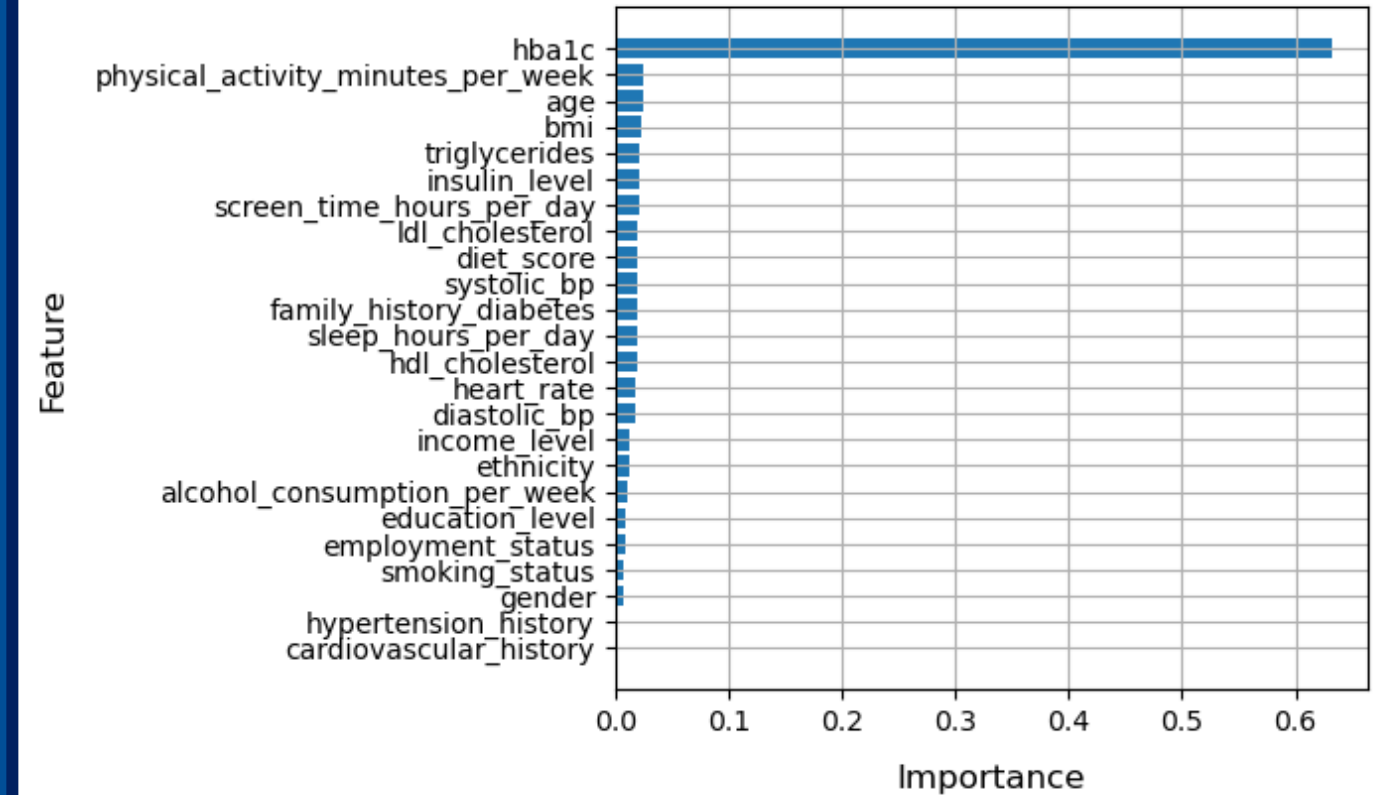
Feature Selection

Random Forest Feature Screening

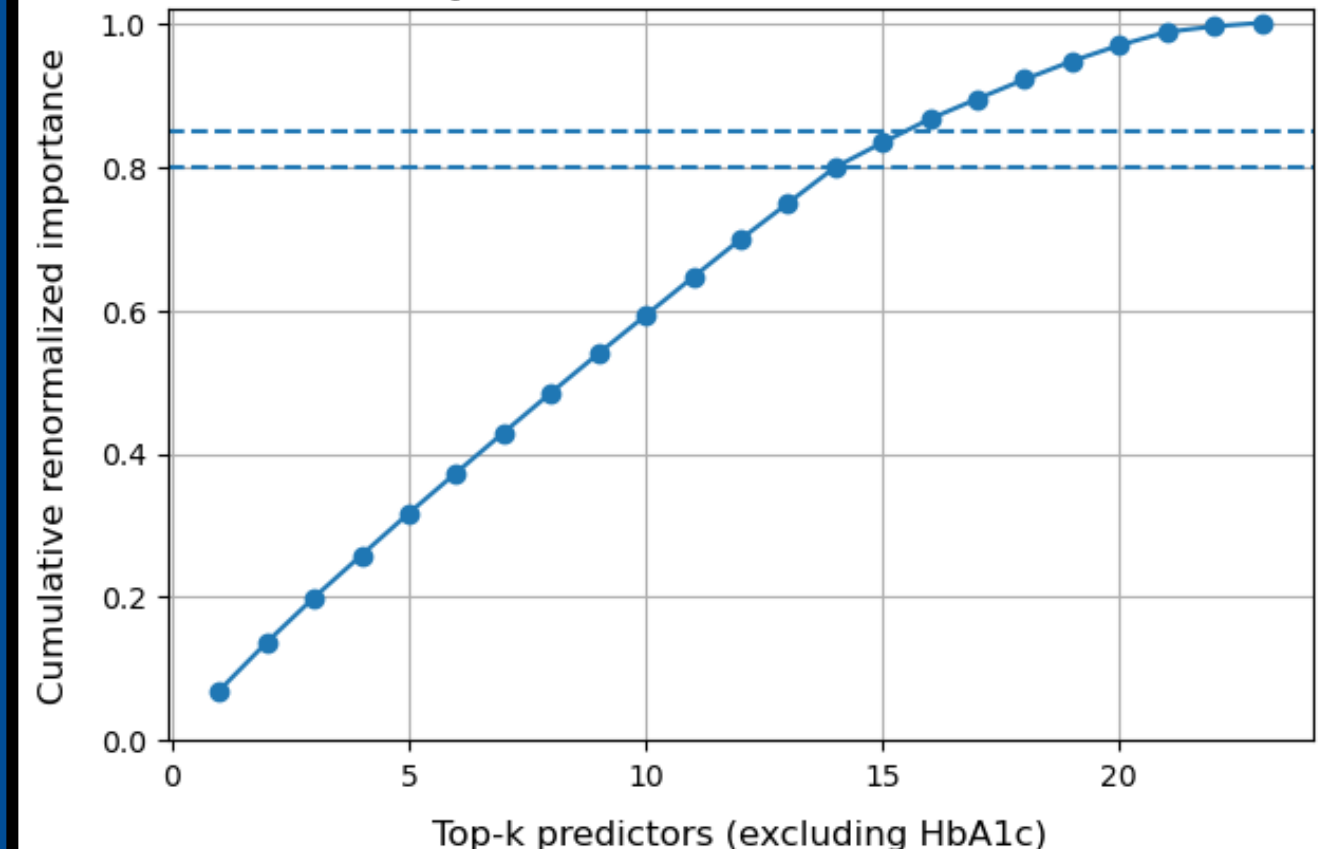
RF Feature Importance - HbA1c Excluded



Aggregated RF Feature Importance (MDI)



Cumulative RF Importance - HbA1c Excluded (Renormalized)



Full vs Reduced

Comparison of models implements via a Likelihood Ratio Test (LRT),

$$LR = 2(l_{full} - l_{red}), \quad LR \sim X^2_{\Delta df}$$

METRIC	FULL MODEL	REDUCED MODEL
Log Likelihood	-27964.2971	-27975.7443
AIC	56002.5942	55983.4887
BIC	56346.3161	56132.1252

STATISTIC	VALUE
LR Statistic	22.8945
ΔDegrees of Freedom	21
p-value	0.3496

Based on the AIC, BIC, and LRT results, we move forward with the reduced model.

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Modeling Approaches

Modeling Process

Model Variants Evaluated

- “ALL” Predictors
- “HbA1c” Only
- “Non-HbA1c” (all predictors w/out HbA1c)

Metrics

Threshold-Based:

- Log-loss
- Accuracy
- Precision
- Recall
- F1-Score

Threshold-Free:

- ROC-AUC

Threshold Evaluation

Youden's-J:

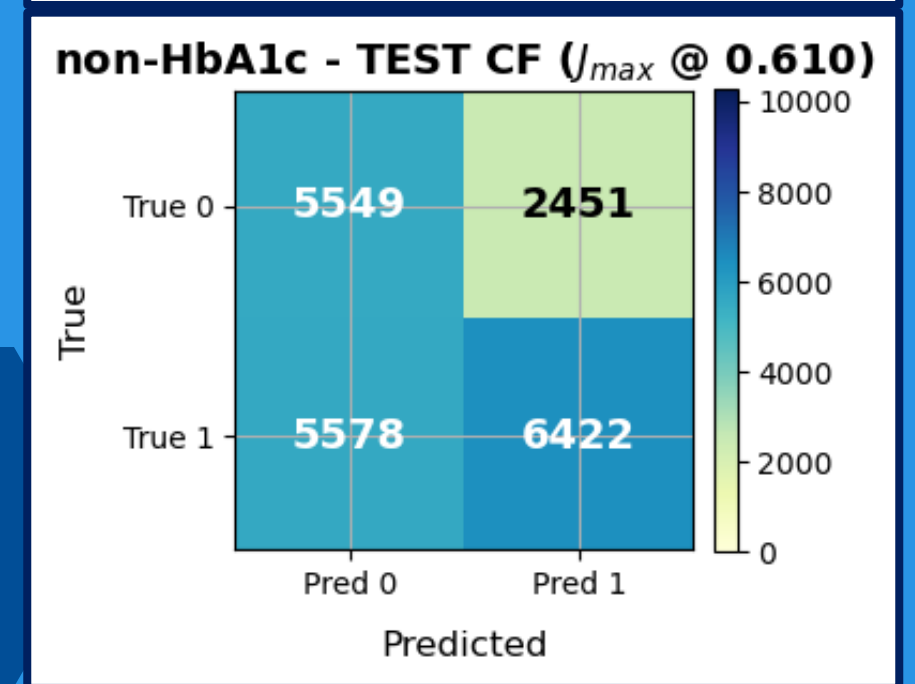
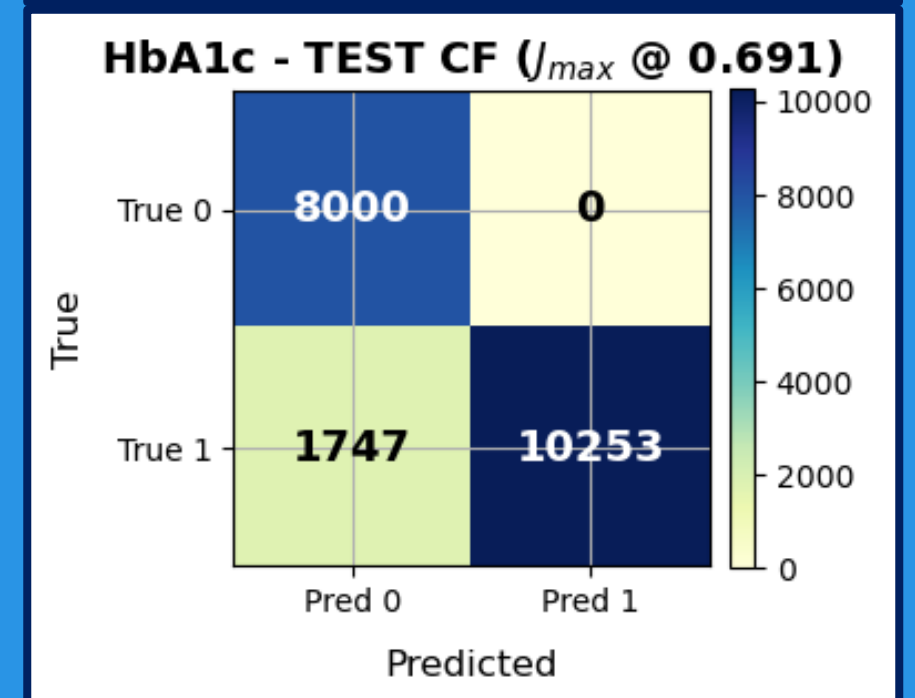
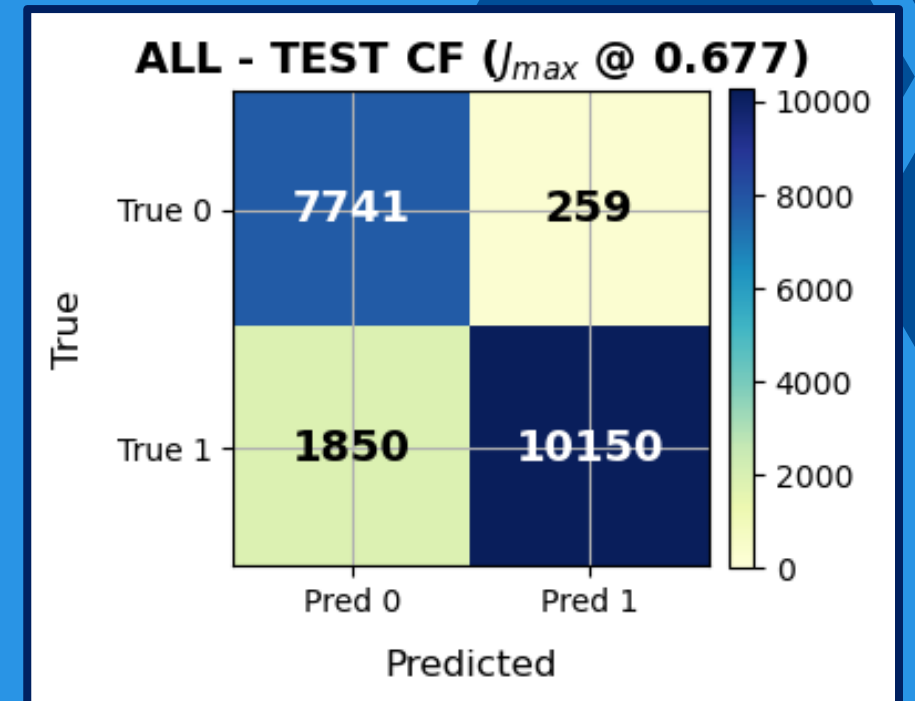
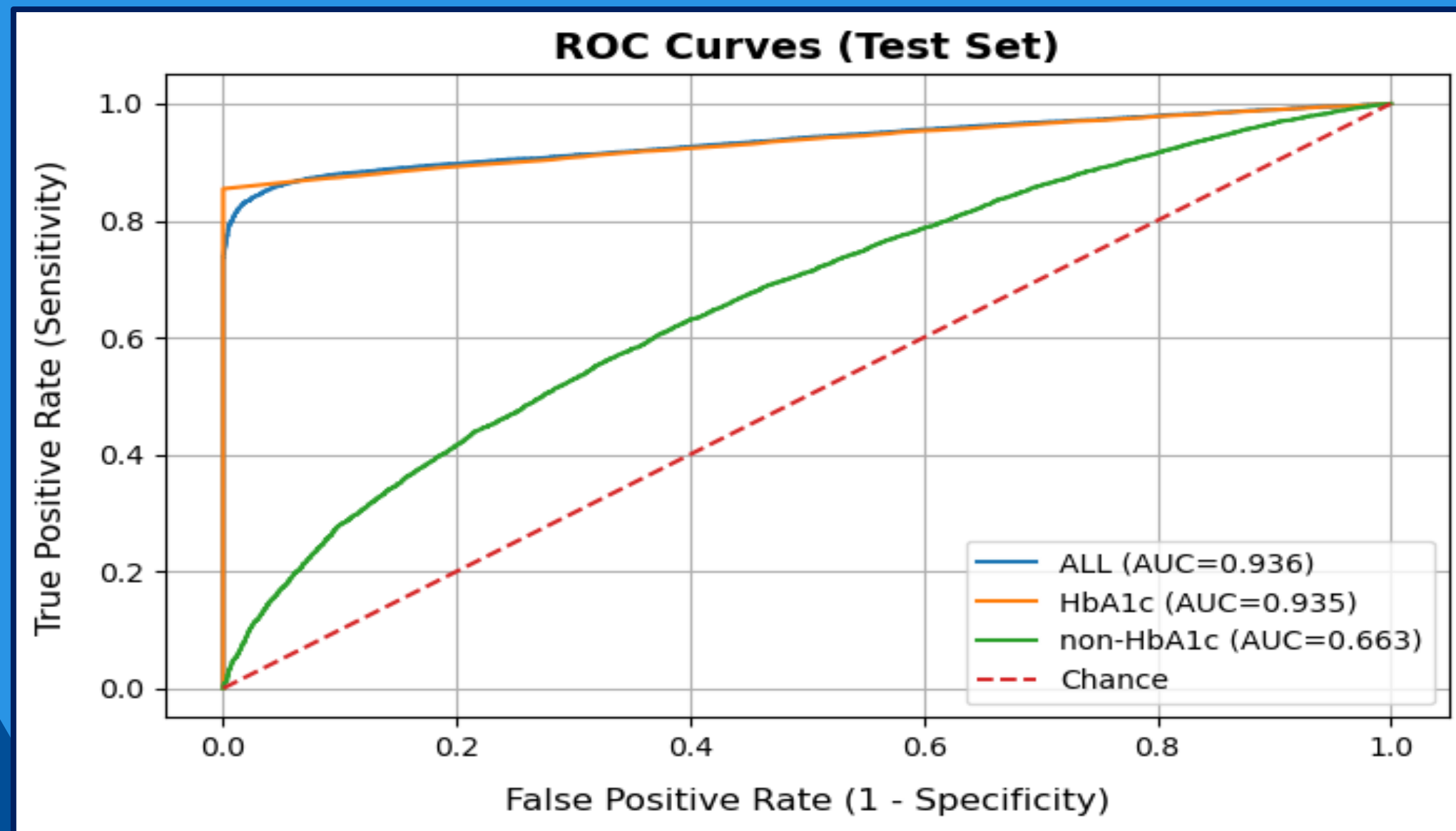
$$J = \text{sensitivity} + \text{specificity} - 1$$

$$J = \frac{TP}{TP + FN} + \frac{TN}{TN + FP} - 1 = \frac{TP \times TN - FP \times FN}{(TP + FN)(TN + FP)}$$

- Maximizes True Positive Ratio to False Positive Ratio on the ROC.
- Balances Sensitivity & Specificity & is independent of class prevalence.

Logistic Regression

MODEL	THRESHOLD	LOG-LOSS	ACCURACY	PRECISION	RECALL	F1-SCORE
ALL	0.6773	0.3409	0.8946	0.9751	0.8458	0.9059
HbA1c – only	0.6915	0.3449	0.9127	1.0000	0.8544	0.9215
Non – HbA1c	0.6103	0.6315	0.5986	0.7238	0.5352	0.6153



LogReg – Predictor Effects

ALL-Predictors Variant

VARIABLE	COEF	<i>p</i> -value
HbA1c	3.5558	0.0000
Family History of Diabetes (Y/N)	0.4986	0.0000
Physical Activity (min/week)	-0.0013	0.0000
Age	0.0079	0.0000
BMI	0.0138	0.0002
HDL Cholesterol	-0.0038	0.0006
Diet Score	-0.0154	0.0116
Triglycerides	0.0007	0.0127
...	...	...

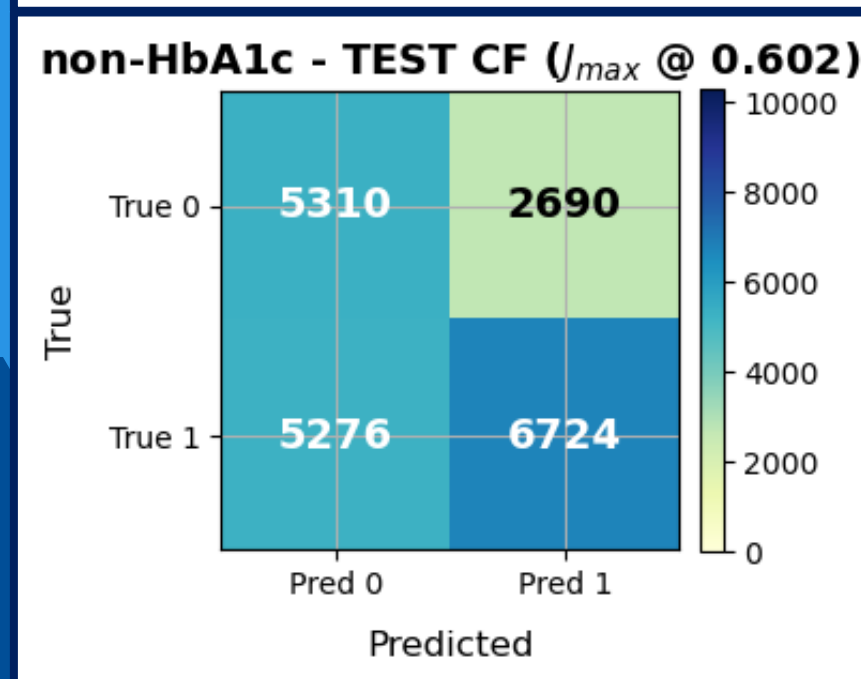
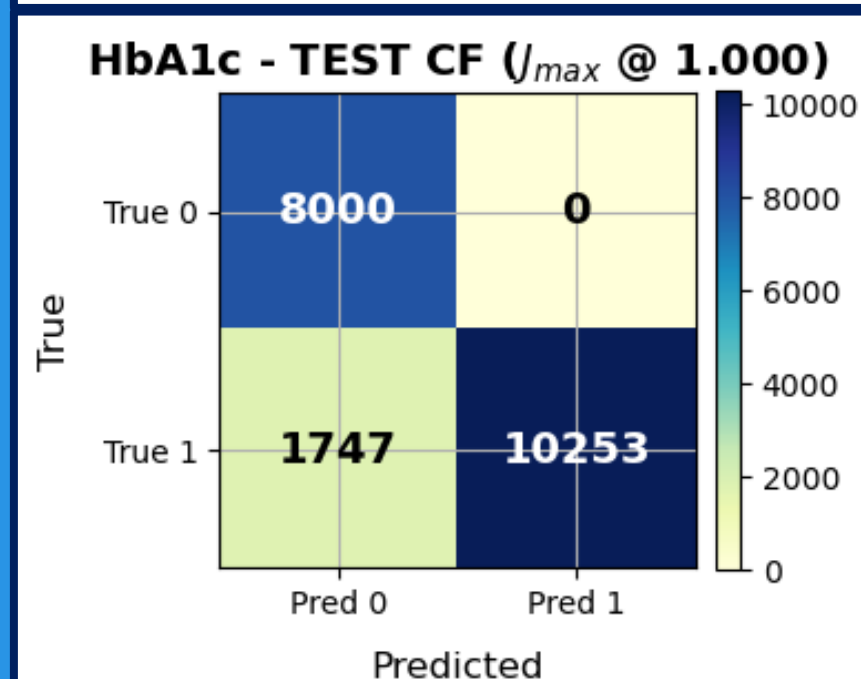
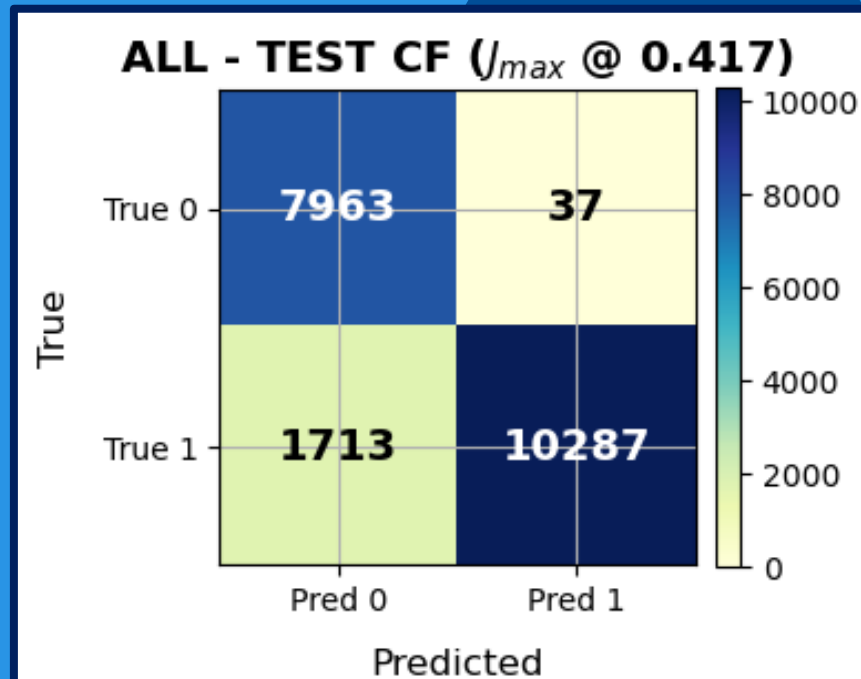
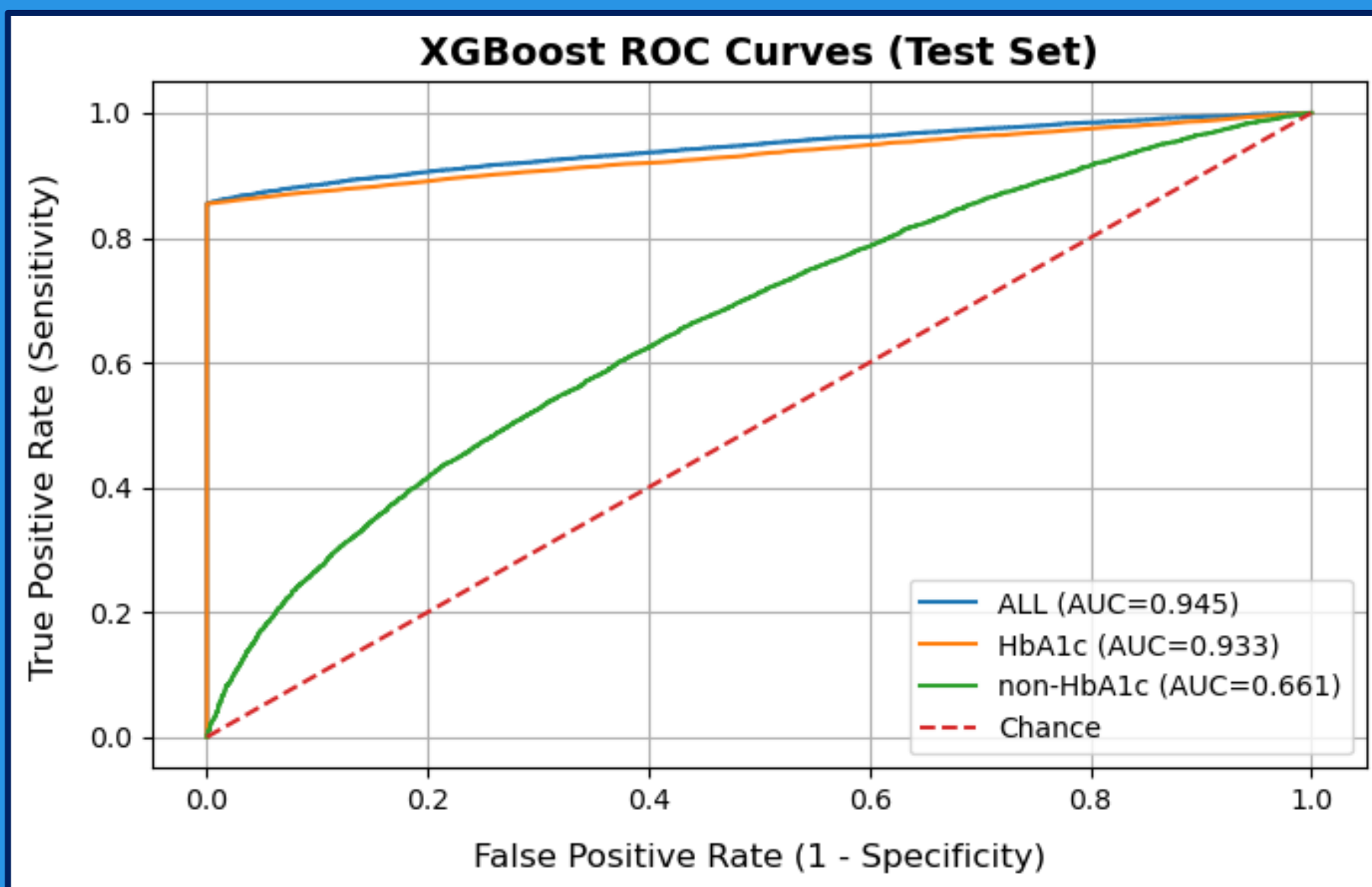
Non-HbAlc Variant

VARIABLE	COEF	<i>p</i> -value
Family History of Diabetes (Y/N)	1.1240	0.0000
Age	0.0176	0.0000
Physical Activity (min/week)	-0.0025	0.0000
BMI	0.0290	0.0000
Diet Score	-0.0380	0.0000
HDL Cholesterol	-0.0065	0.0000
Insulin Level	0.0093	0.0000
Triglycerides	0.0011	0.0000
Screen Time/Day	0.0148	0.0000
Systolic BP	0.0016	0.0133
...	...	...

NOTE: Tables show significant predictors per variant ($p < 0.05$).

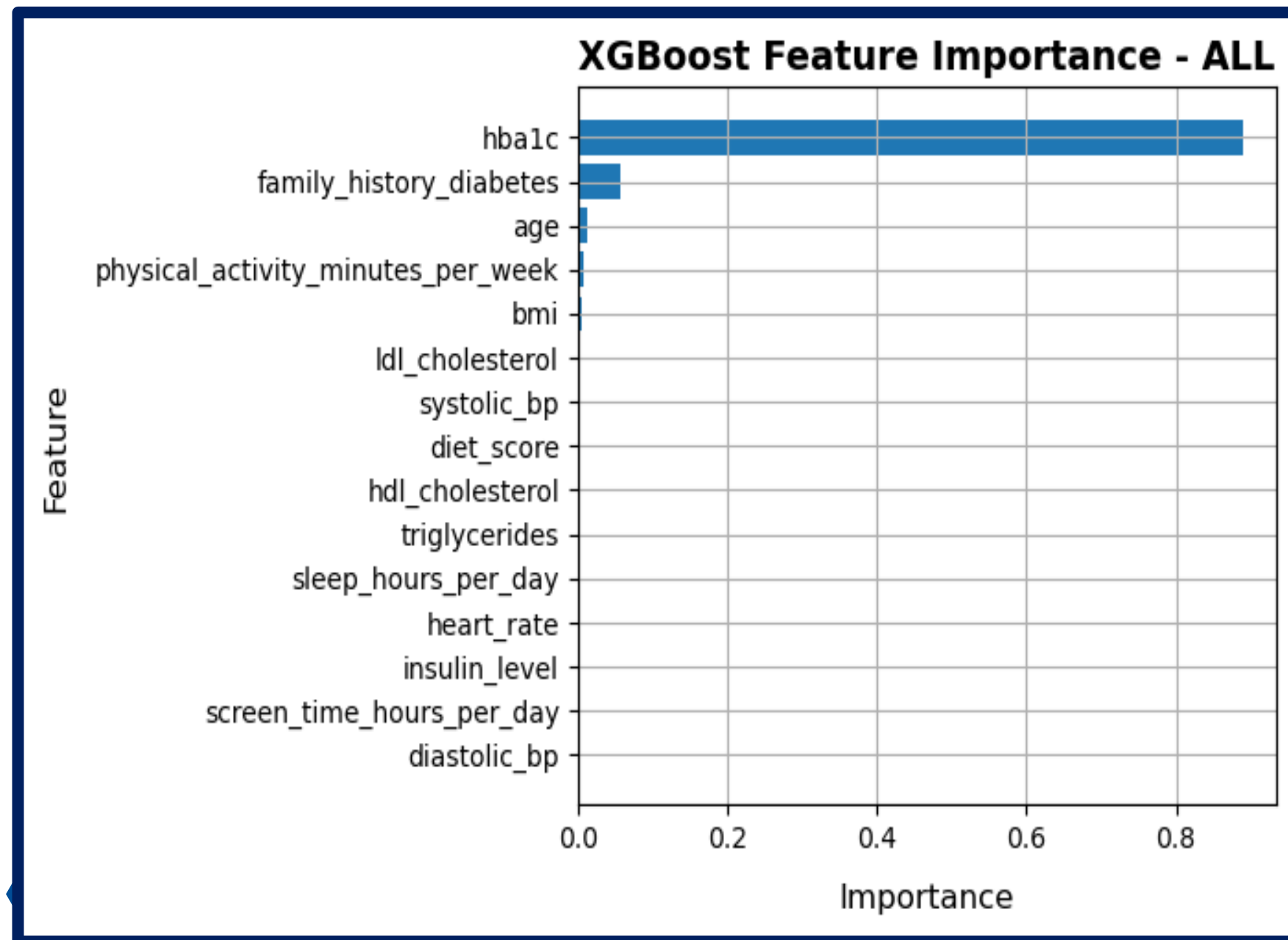
XGBoost

MODEL	THRESHOLD	LOG-LOSS	ACCURACY	PRECISION	RECALL	F1-SCORE
ALL	0.4166	0.2224	0.9125	0.9964	0.8573	0.9216
HbA1c – only	0.9999	0.2287	0.9127	1.0000	0.8544	0.9215
Non – HbA1c	0.6015	0.6325	0.6017	0.7143	0.5603	0.6280

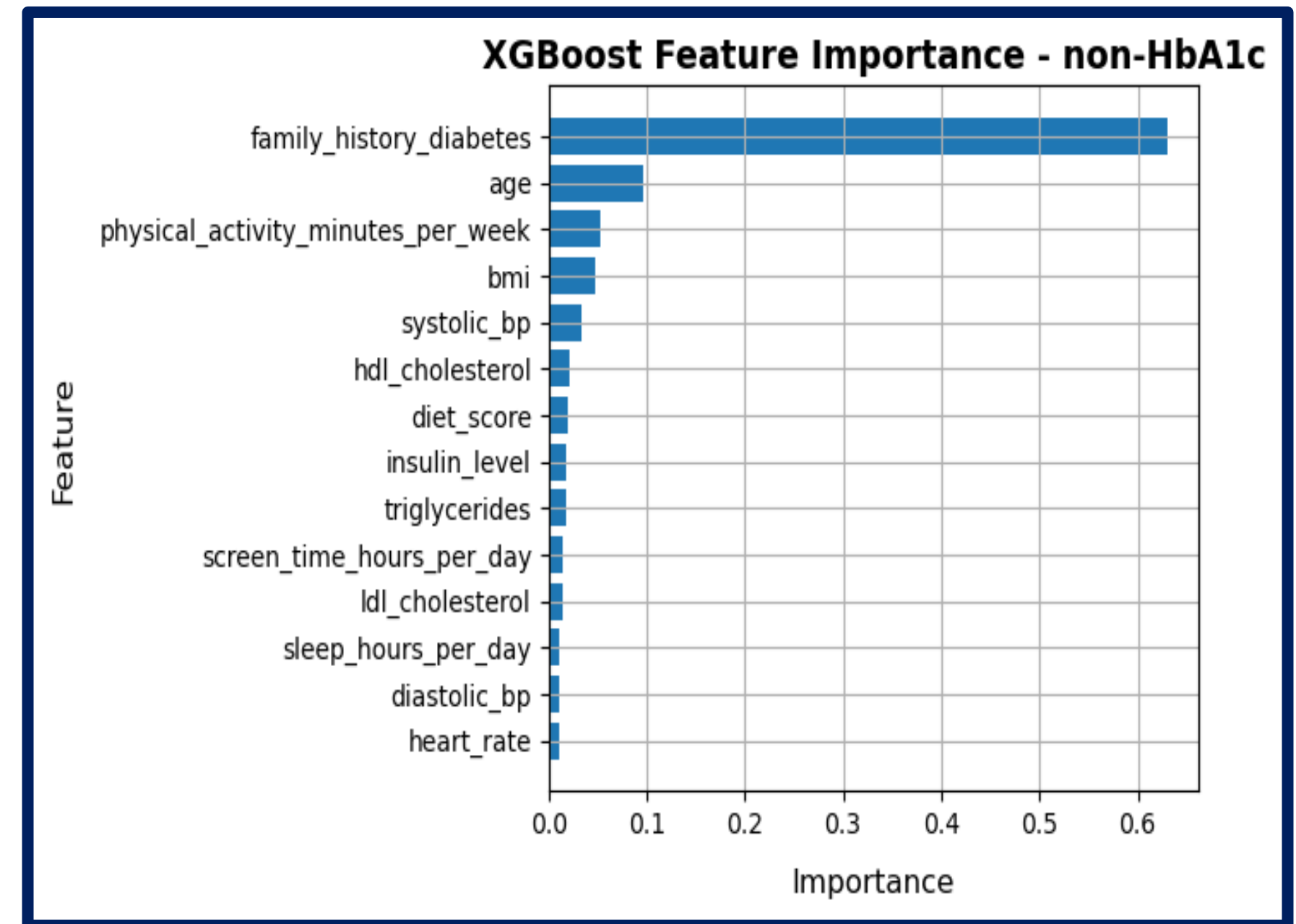


XGBoost – Predictor Effects

ALL-Predictors Variant

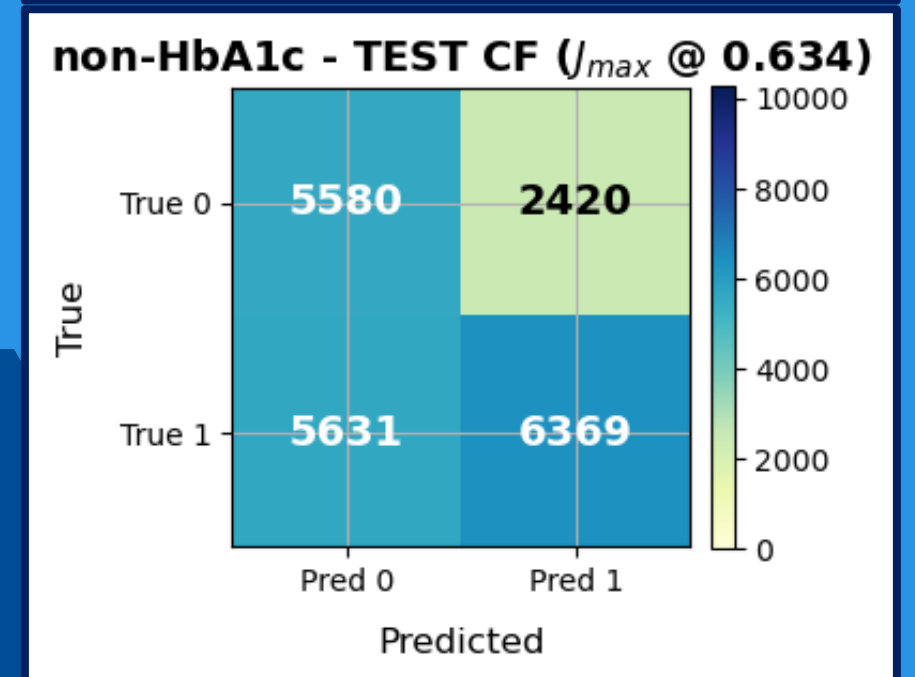
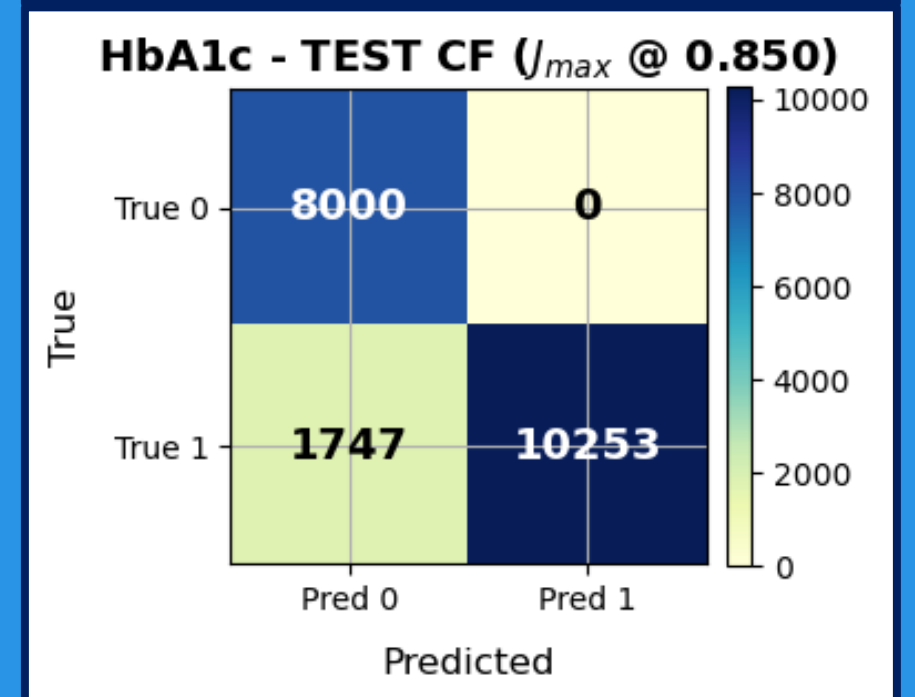
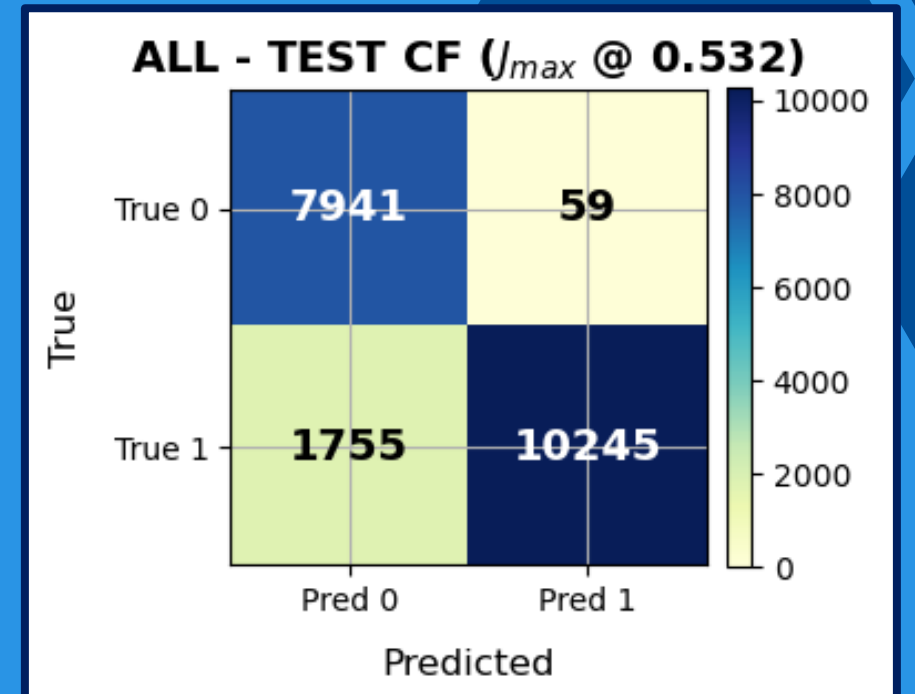
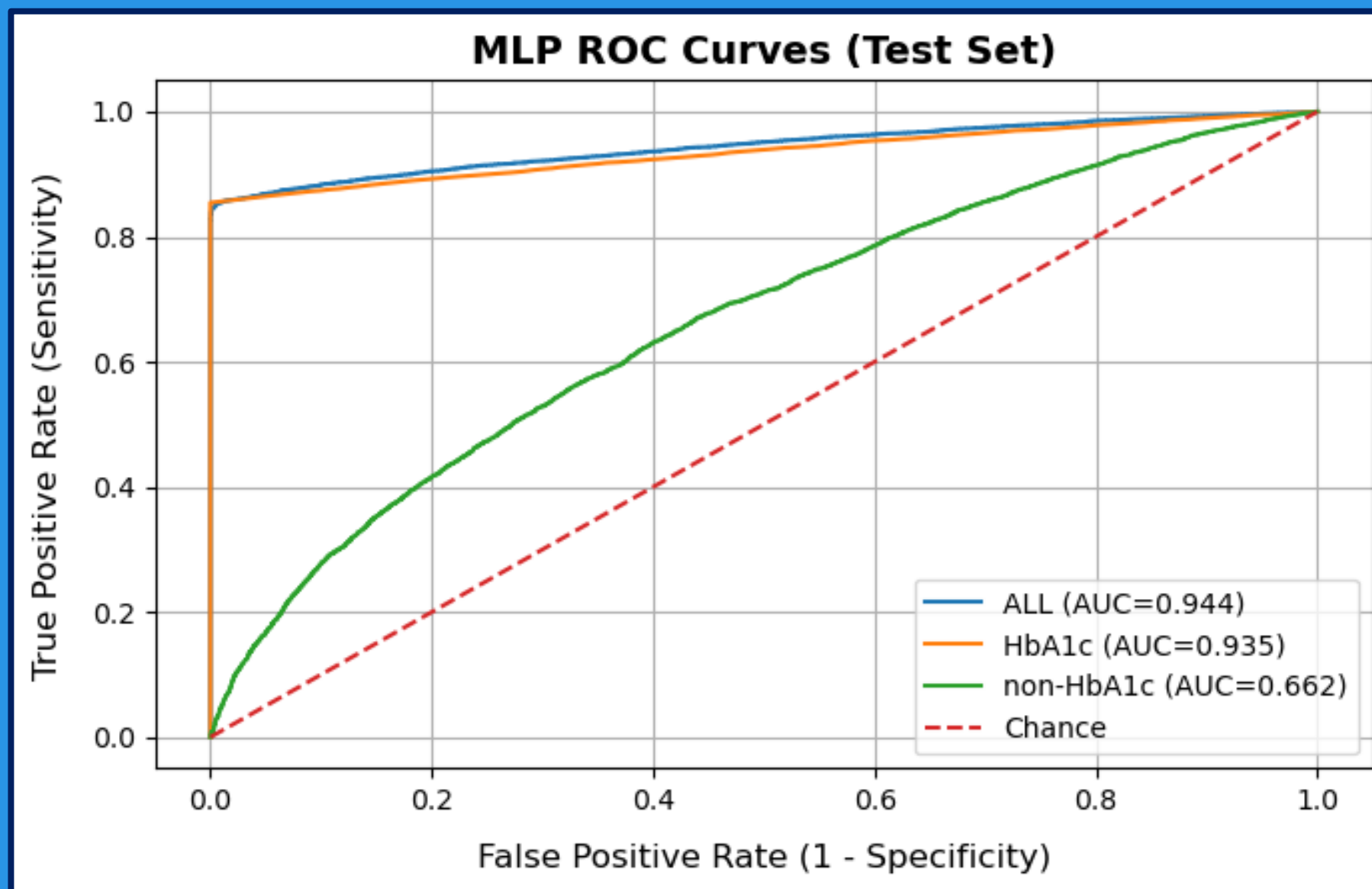


Non-HbA1c Variant



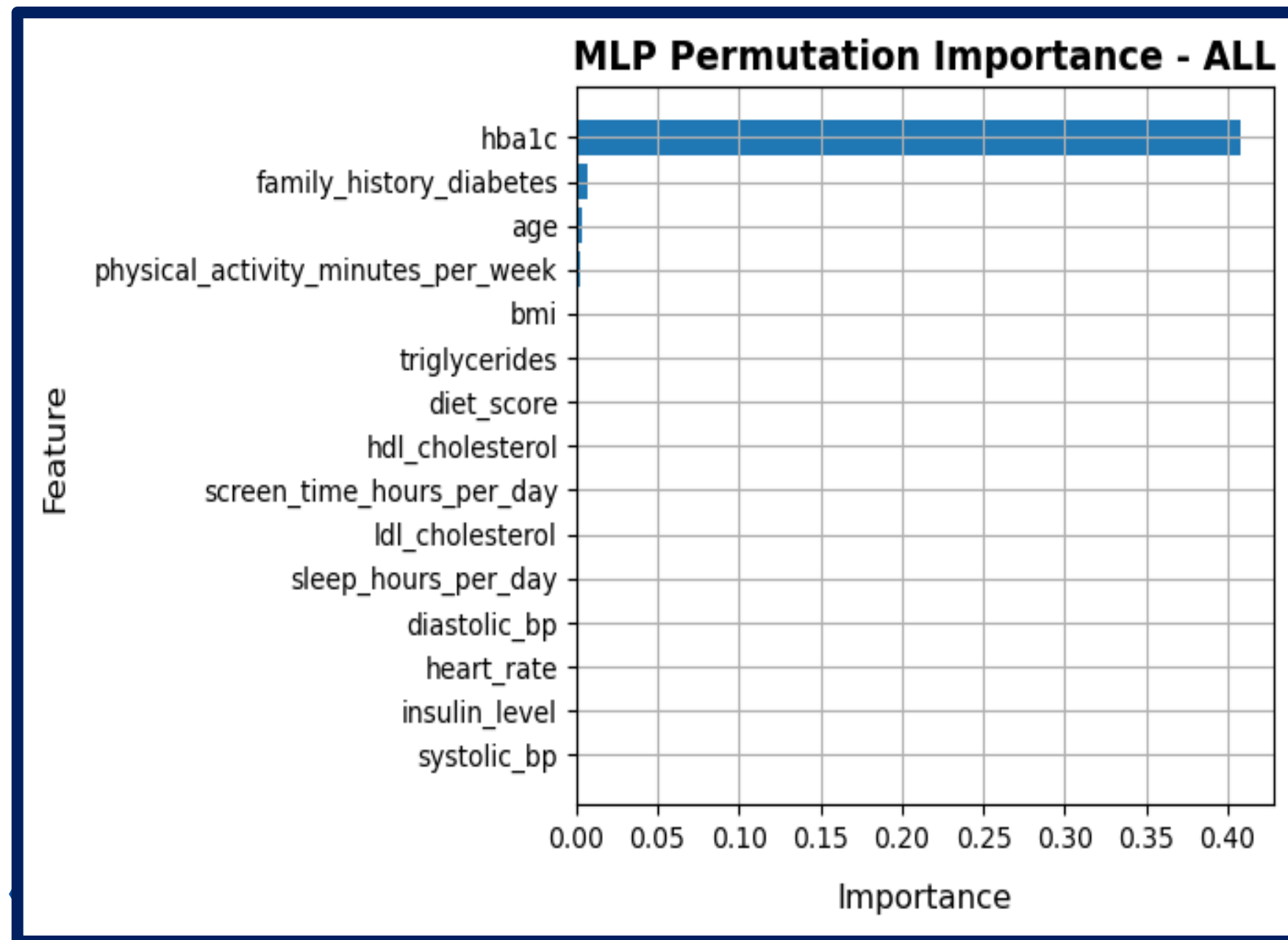
Multi-Layer Perceptron

MODEL	THRESHOLD	LOG-LOSS	ACCURACY	PRECISION	RECALL	F1-SCORE
ALL	0.5324	0.2426	0.9093	0.9943	0.8538	0.9187
HbA1c – only	0.8501	0.2446	0.9127	1.0000	0.8544	0.9215
Non – HbA1c	0.6337	0.6329	0.5975	0.7247	0.5308	0.6127

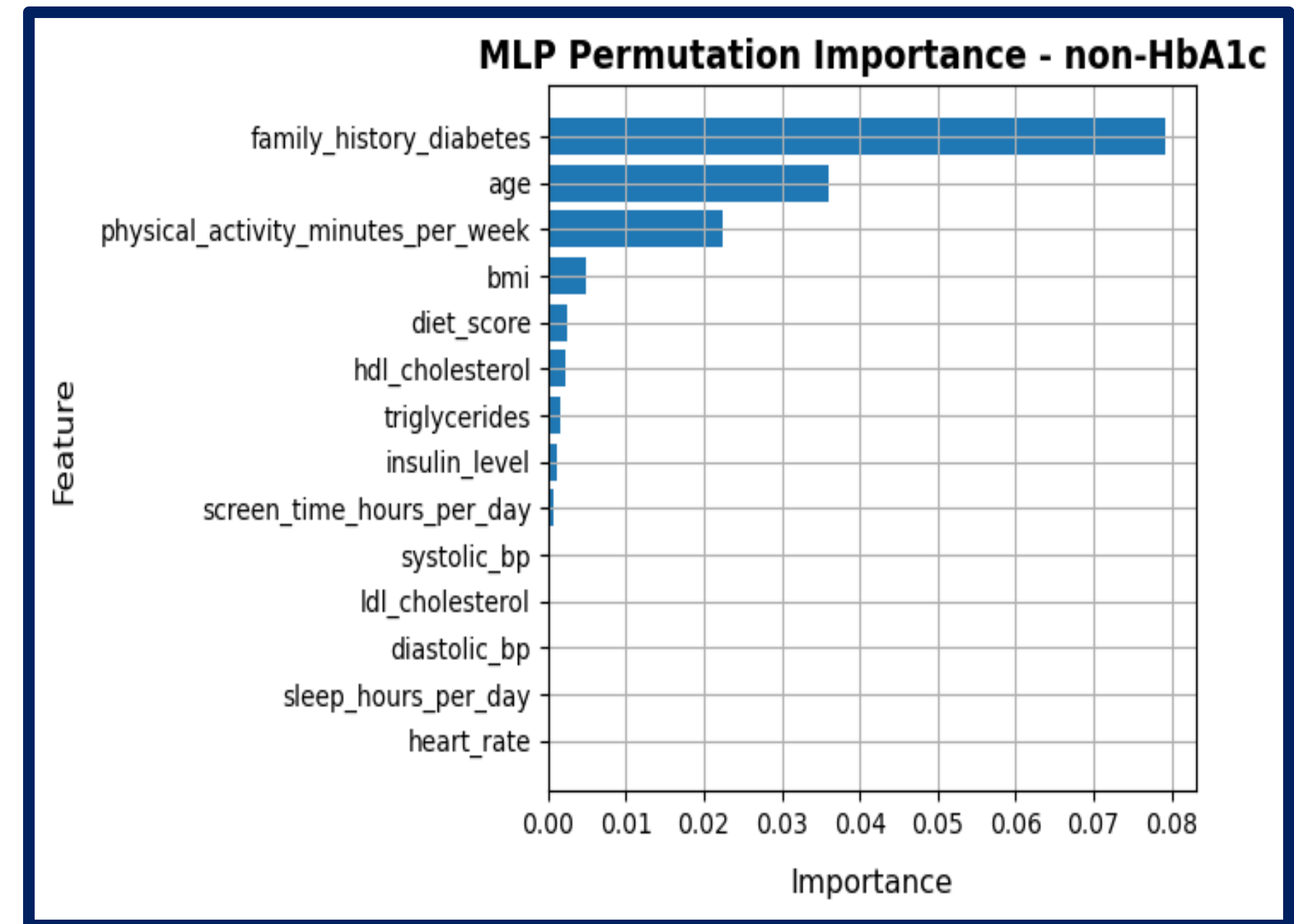


MLP – Predictor Effects

ALL-Predictors Variant



Non-HbA1c Variant



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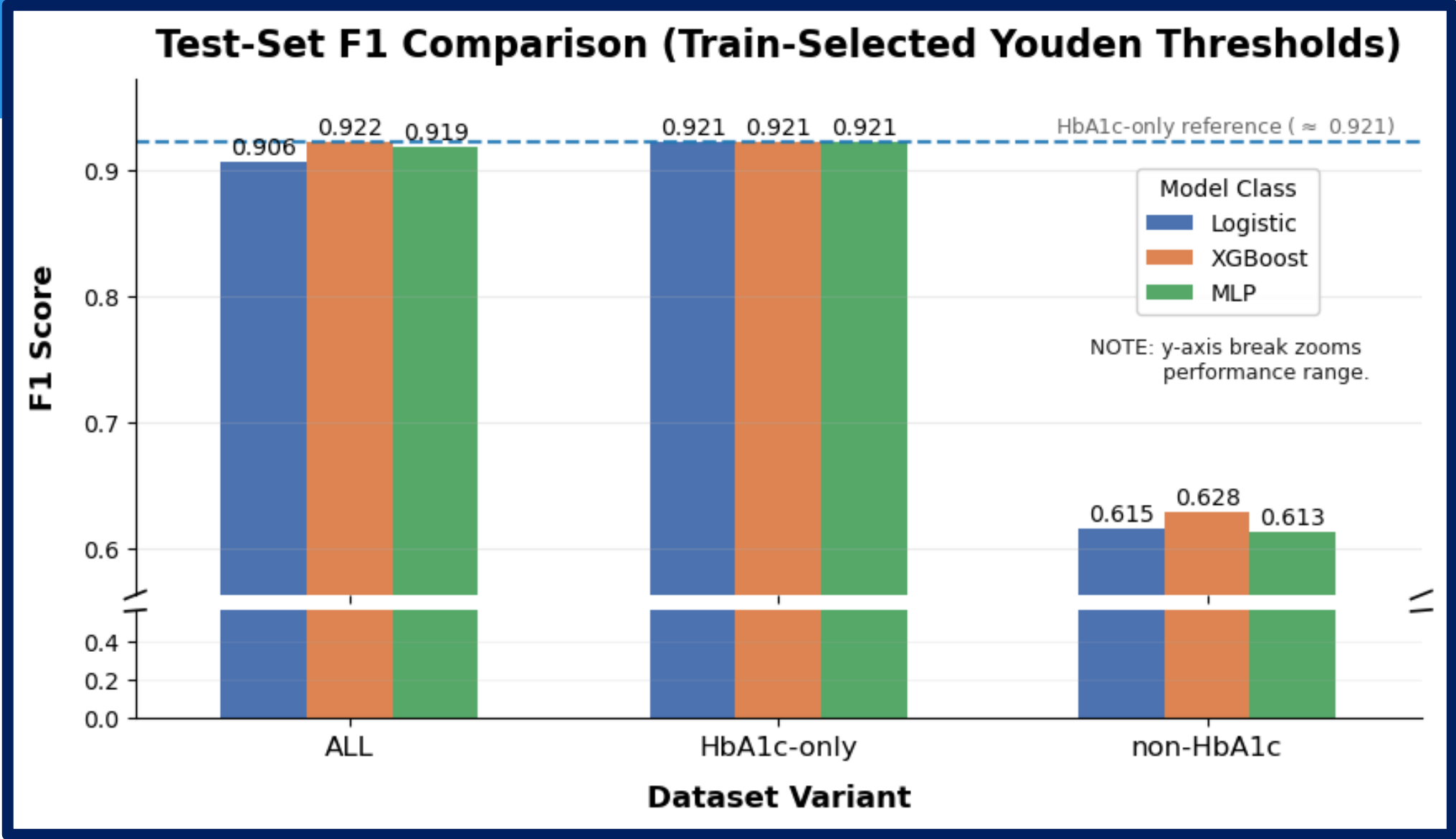
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Cross-Model Comparisons

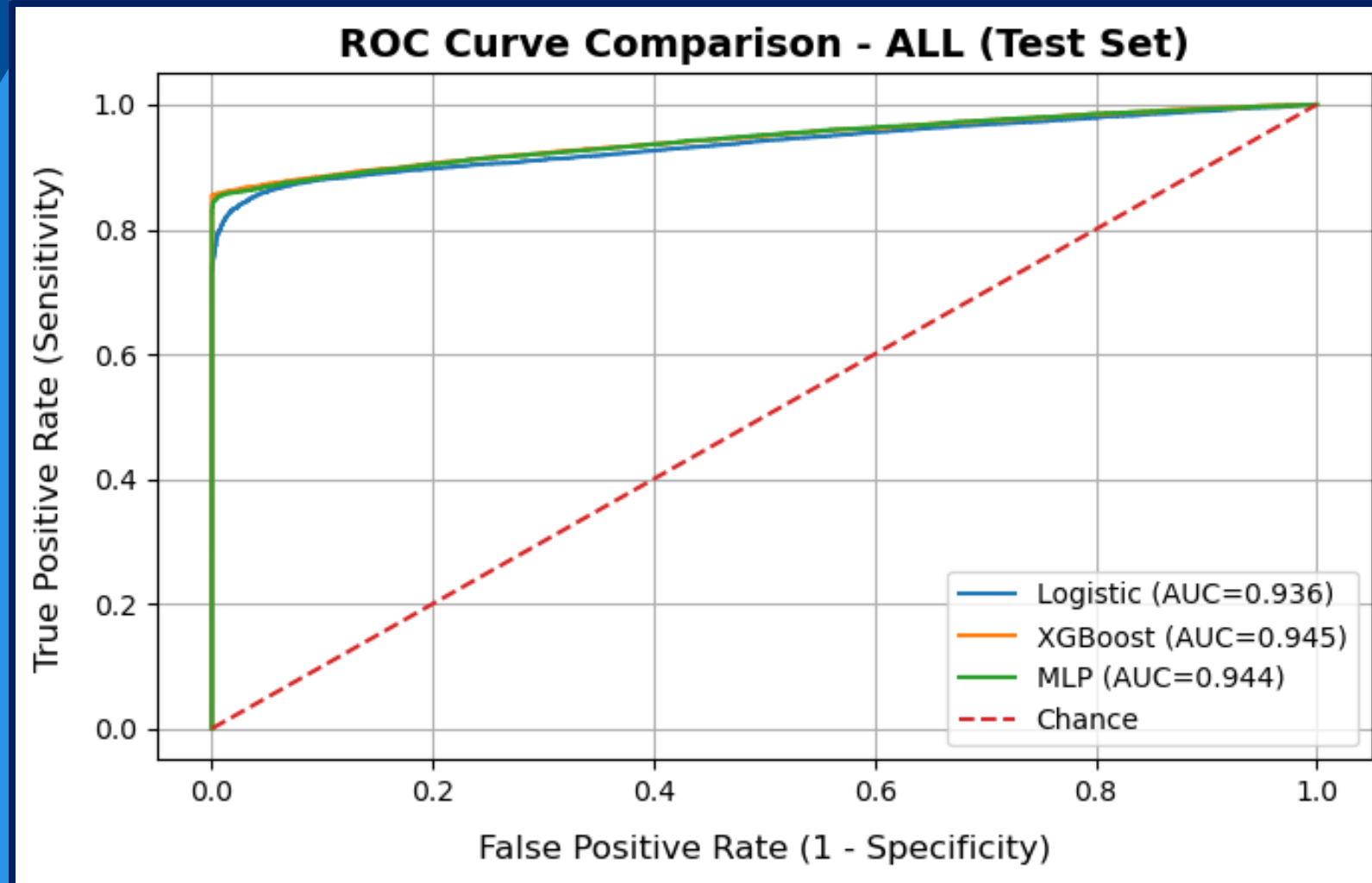
Performance Across Models

Test-Set Performance (Train-Selected Youden Threshold)



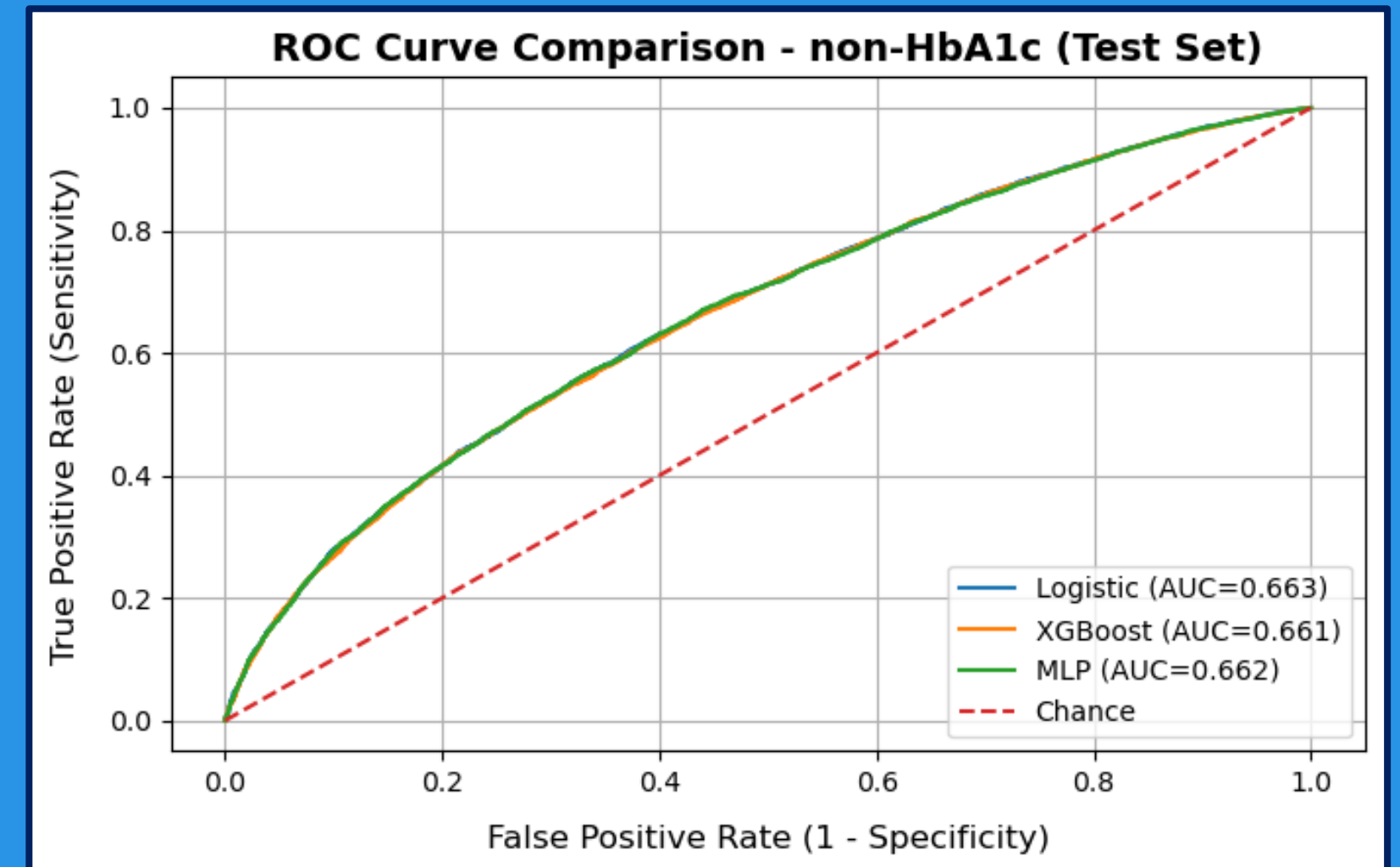
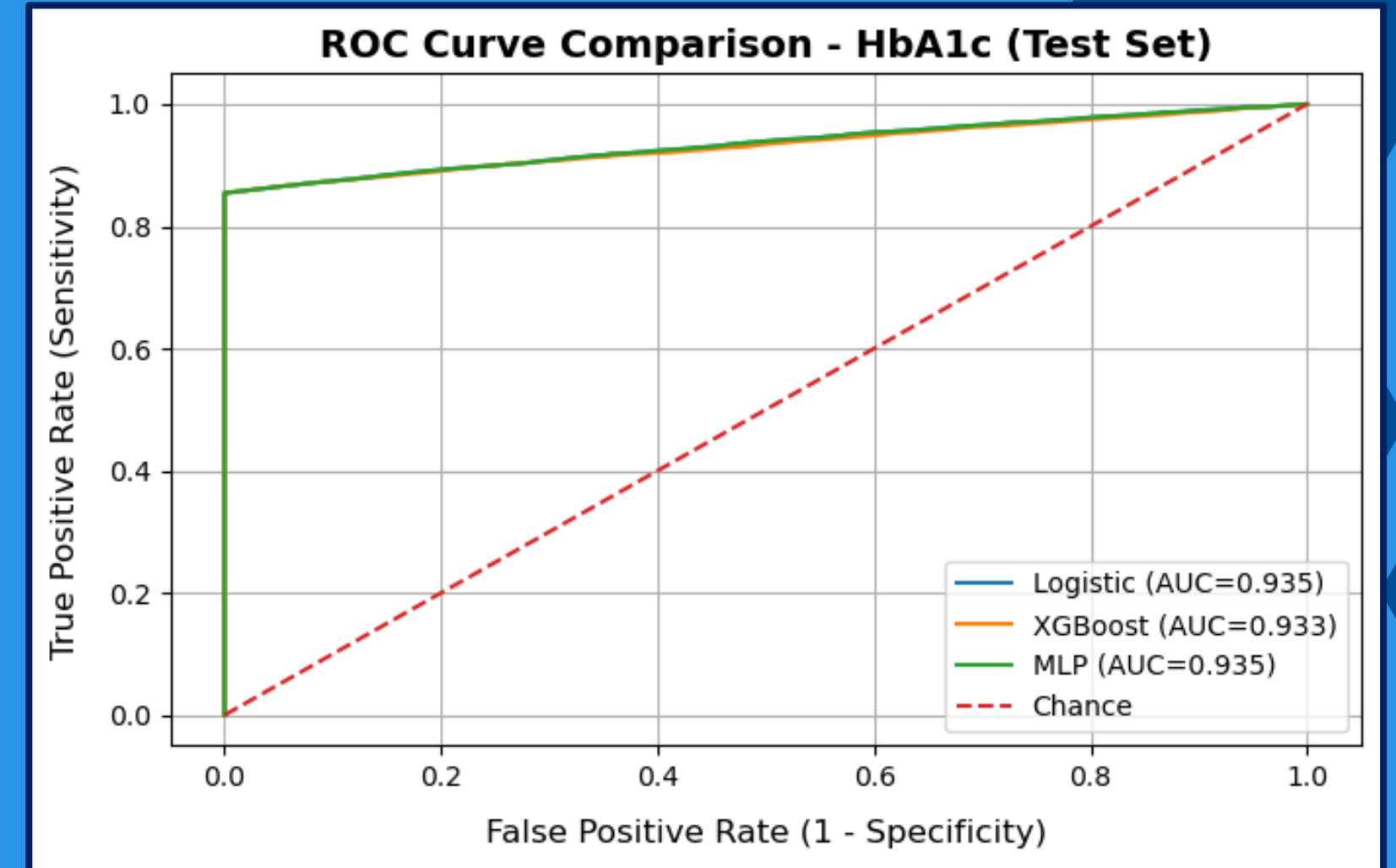
VARIANT	MODELING METHOD	ACCURACY	PRECISION	RECALL
ALL	Logistic	0.8946	0.9751	0.8458
	XGBoost	0.9125	0.9964	0.8572
	MLP	0.9093	0.9943	0.8538
HbA1c-only	Logistic	0.9126	1.0000	0.8544
	XGBoost	0.9126	1.0000	0.8544
	MLP	0.9126	1.0000	0.8544
Non-Hba1C	Logistic	0.5986	0.7238	0.5352
	XGBoost	0.6017	0.7143	0.5603
	MLP	0.5974	0.7247	0.5308

ROC Comparison



- HbA1c is the dominant predictor of diabetes diagnosis.
- Removing HbA1c substantially reduces model performance.
- Differences between modeling methods are relatively small.

Feature Choice impacts performance more than Model Choice.



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Key Findings

Key Findings – Research Questions Revisited

Q1 (Primary): Model Comparison

- Logistic Regression, XGBoost, and MLP produced **similar predictive performance**.
- Increased model complexity provided **little to no advantage**.

Q2 (Secondary): Influential Predictors

- **HbA1c** emerged as the **dominant predictor** across all interpretability methods.
- Secondary predictors included **family history of diabetes, age, physical activity, and BMI**.

Q3 (Secondary): Incremental Value of HbA1c

- **HbA1c-only models** performed nearly as well as full models.
- Removing HbA1c caused a **substantial decline in performance**.

Feature choice mattered more than model complexity.

HbA1c captured most of the predictive signal.

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Limitations & Future Directions

Study Limitations

- **Synthetic dataset:** May not fully capture the complexity of real clinical data
- **Dataset Documentation Inconsistencies:** Additional preprocessing assumptions
- **Binary outcome modeling:** Combined multiple stages into one diagnosis class
- **Cross-sectional predictors:** Does not capture longitudinal disease progression
- Findings **may not fully generalize** to other clinical populations or healthcare settings

These limitations highlight that our goal was primarily to **compare modeling approaches and predictor patterns**, rather than to produce a deployable clinical model.

Future Directions

Alternative Target Variables:

- Model more-granular target, **diabetes stage**, to explore multi-class or ordinal classification
- Investigate **diabetes risk score** for early risk prediction.

Examine non-HbA1c Predictors:

- Current method is *ablation study*; explore modeling methods omitting HbA1c **prior to any feature selection decisions**.
- Examine use of **non-biomarker features** when laboratory data are unavailable.

Modeling Approaches:

- Calibration / Hyperparameter tuning
- Regularization Techniques
- Alternative Feature Engineering

Validation of Results:

- Validate results using **real-world clinical datasets**.



Thank you!

Any Questions?